

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Campus Lost & Found Management System

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Abstract

The Campus Lost & Found Management System is a mobile application developed to make reporting and recovering lost items easier within a college campus. The application allows students to create accounts, report lost or found items, provide details such as item name, description, category, location, and date, and attach item photos. Firebase Authentication is used for secure user login and registration, while Cloud Firestore is used to store and manage item reports.

The system includes a matching feature that compares lost and found item details such as item name, description, category, location, date, and image information to identify possible matches. Users can view matching items and proceed with the claim and verification process. The application is developed using Kotlin and Jetpack Compose with Firebase services for backend functionality. Overall, the system provides a simple, organized, and efficient platform for managing lost and found items on campus and helps reduce the time required to recover misplaced belongings.

Problem Statement

PROBLEM STATEMENT:

Finding lost items in a college campus can be difficult because there is often no single system to report and manage them. Students usually depend on notice boards, WhatsApp groups, or asking others, which can make the process slow and confusing. Found items may also remain unclaimed because students do not know who the owner is.

The proposed Campus Lost & Found Management System provides a single platform where students can report lost and found items, add important details and photos, and search for possible matches. The system helps organize all reports in one place and makes it easier for students to communicate and recover their belongings.

Objectives

OBJECTIVES:

- To develop a mobile application for reporting lost and found items on campus.
- To allow students to add item details such as name, description, category, location, date, and photo.
- To securely store and manage lost and found item records.
- To compare lost and found reports and identify possible matching items.
- To provide a simple way for users to view and claim recovered items.
- To reduce the time and effort needed to find lost belongings.
- To provide administrators with better control over lost and found records.

Existing Systems / Website :

- Web-based lost & found reporting
- Manual search of reported items
- Detailed forms and supporting documents
- Basic search/filter facilities
- Ownership/identity verification workflow
- Recovery/claim process
- Contact/notification features

Existing Campus Lost & Found University Portal
lostandfoundcampusapp.pythonanywhere.com

The existing portal supports reporting, searching, verification and recovery of lost items. Our Android system adds automated, explainable matching using multiple item attributes and a weighted scoring method.

Our Proposed System :

- Android mobile application
- Automated item matching
- Structured item reporting with essential details
- Multi-factor comparison of name, description, category, location & date
- Match score + confidence + explanation
- Possible-match detection before claiming
- Firebase-based authentication & data storage

Research Gap

Existing lost and found systems mainly focus on reporting, searching, verification, and recovery of items. From the systems reviewed, there is limited emphasis on automatically comparing lost and found reports using multiple attributes such as item name, description, category, location, and date. This creates an opportunity for a campus-focused system that combines structured reporting with automated and explainable item matching. Our proposed system addresses this gap by generating match scores, confidence levels, and matching explanations to help users identify possible matches more efficiently.

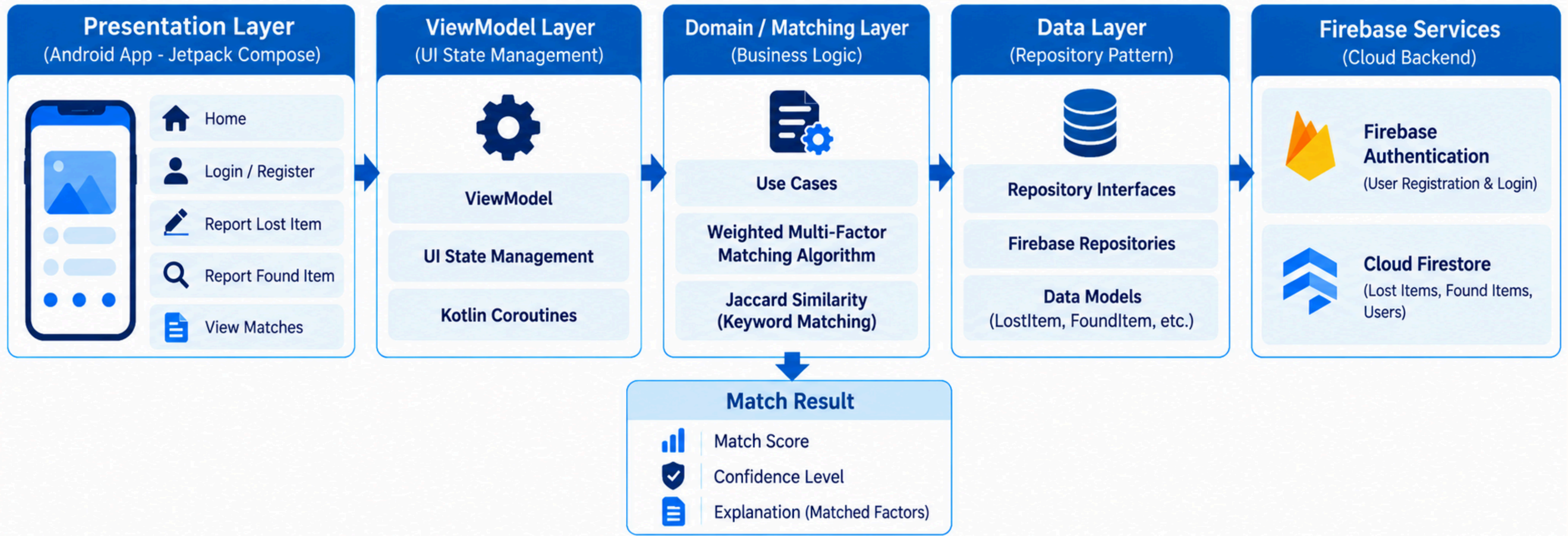
Proposed Methodology

The proposed Campus Lost & Found Item Matching System is developed as an Android mobile application that provides a centralized platform for managing lost and found items on campus. Users can securely register and log in using Firebase Authentication. After logging in, users can report a lost or found item by entering details such as item name, description, category, location, date, and photo. These records are stored in Cloud Firestore for easy access and management.

The system compares lost and found reports using a weighted multi-factor matching algorithm. It uses Jaccard similarity for keyword comparison and evaluates item name, description, category, location, and date proximity to generate a match score, confidence level, matched factors, and explanation. Possible matches are then displayed to the user. The application is developed using Kotlin, Jetpack Compose, Firebase Authentication, and Cloud Firestore.

System Architecture

Campus Lost & Found Item Matching System



Technologies Used

- **React Native** - Development of the mobile application UI
- **JavaScript / TypeScript** - Programming language for the application
- **Firebase Authentication** - User registration and login
- **Cloud Firestore** - Storage and management of lost/found item data
- **Firebase Cloud Messaging (FCM)** - Sending notifications to users
- **Firebase Storage** - Storing item images when enabled
- **Google ML Kit** - Image labeling and extracting useful image information
- **Matching Algorithm** - Comparing item name, description, category, location, date, and image information
- **Expo** -Development and testing environment for React Native
- **Android Studio / Android Emulator** - Android testing and debugging
- **Git & GitHub** -Version control and project management

Dataset / Input Details

The system will use user-generated campus lost and found reports as its main dataset. Each report is stored in Cloud Firestore and contains information needed for searching and matchin

Input Details :

- **Item Name:** Name of the lost or found item.
- **Description:** Details such as color, brand, size, or unique features.
- **Category:** Wallet, phone, bag, books, ID card, electronics, etc.
- **Location:** Place where the item was lost or found.
- **Date:** Date the item was lost or found.
- **Image:** Photo of the item, when available.
- **User ID:** Identifies the student who submitted the report.
- **Status:** Lost, Found, Claimed, or Resolved.

The collected information is used by the matching algorithm to compare lost and found reports using keywords, category, location, date, and image-related information.

Implementation – Module 1

Implementation – Module 1: User Authentication

Module 1 focuses on implementing user registration, login, and logout for the Campus Lost & Found Item Matching System.

The application uses Firebase Authentication with the Email/Password method. A registration screen allows new users to create an account using their email address and password. During login, Firebase verifies the entered credentials and grants access to the application when they are valid.

After successful authentication, the user is redirected to the home dashboard, where they can report lost items, report found items, and view possible matches. The application also maintains the user's authentication state so that a logged-in user does not need to sign in repeatedly.

The module also includes basic input validation and error handling for invalid credentials, empty fields, and registration failures. This provides a secure foundation for the remaining modules of the system.

Implementation – Module 2

Module 2 focuses on implementing the Lost and Found Item Reporting functionality. Users can report a lost or found item by entering details such as item name, description, category, location, and date. They can also select an image of the item from their device. The submitted information is validated before being stored in Cloud Firestore. Separate collections are maintained for `lost_items` and `found_items`, allowing the system to organize reports efficiently. Each report is associated with the user's ID and includes a status such as Lost or Found. This module provides the basic dataset required for the matching system. The collected item details can later be compared to identify possible matches between lost and found reports.

Implementation – Module 3

Module 3 focuses on implementing the item matching system. The system compares the details of lost and found items to identify possible matches.

The matching process considers factors such as item name, description, category, location, and date. Keyword similarity is used to identify common words between lost and found item descriptions, while category and location matches improve the overall score.

A weighted scoring method is used to calculate a Match Score for each possible pair. The results are sorted based on the highest score and displayed to the user as possible matches. This helps users quickly identify found items that may belong to them and reduces the need for manual searching.

Experimental Setup

Development Environment :

- Android Studio
- Kotlin
- Jetpack Compose
- Gradle

Testing Environment :

- Pixel 5 Android Emulator
- Android API 34
- Internet connection
- Cloud Firestore test data

Backend / Database :

- Firebase Authentication
- Cloud Firestore

Test Data & Evaluation

- Lost and found item records
- Item name, description, category, location, date
- Search and filter testing
- Matching score and confidence testing
- Login and report-submission testing

Screenshots / Working Demo

Campus Reports 5 [View Matches →](#)



Silver Water Bottle LOST

Stainless steel silver bottle with a black cap and small scratch near the bottom

Computer Lab 15-09-2026

Water Bottle [View Match →](#)

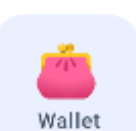


Blue Backpack LOST

Dark blue backpack with a laptop compartment and small front pocket

Canteen 15-09-2026

Bags [View Match →](#)



Black Wallet FOUND

Black leather wallet with college ID

Library 10-09-2026

Wallet [View Match →](#)

Found: Black Wallet

Category: Wallet

Location: Library

Description: Black leather wallet with college ID

Match Score: 100% [HIGH Confidence]

Matched Factors: Item Name (100%), Description (100%), Category (100%), Location (100%), Date Proximity (100%)

Explanation: HIGH confidence (100%): matching category 'Wallet', strong name overlap ('Black Wallet' ↔ 'Black Wallet'), shared description keywords, matching location 'Library', same/adjacent report date. Breakdown: Name 100%, Desc 100%, Category 100%, Location 100%, Date 100%.

CAMPUS RECOVERY

Report Found Item

Provide details of the item you discovered to help reconnect it with its rightful owner.



Item Details

Key identifiers and physical attributes

Item Name *



e.g. Blue Hydro Flask, Apple AirPods Case

Specify the item type, brand, or dominant color

Category *



e.g. Electronics, Bags, Keys, IDs

Helps classify the found item for matching searches

Description *



Mention visible scratches, stickers, condition, where it was placed...

Avoid sensitive identifiers that only the true owner would know

Comparison with Existing System

Feature	Existing Campus Portal	Our Proposed System
 Platform	Web application	Android mobile application
 Reporting	Detailed lost/found forms	Structured mobile reporting
 Search	Manual search & filters	Search + automated matching
 Matching	Basic/unspecified matching	Weighted multi-factor matching
 Comparison	Item details searched manually	Name, description, category, location & date
 Result	Item listings / recovery process	Match score + confidence + explanation
 Database	Web-based backend	Firebase Cloud Firestore
 Authentication	User verification workflow	Firebase Authentication
 Recovery	Claim / verification process	Possible-match detection before recovery



Key Difference: Our system focuses on **automated and explainable matching** of lost and found items using multiple attributes and a weighted scoring algorithm.

Challenges & Solutions

During the development of the Campus Lost & Found Item Matching System, several challenges were encountered. Managing and organizing lost and found records was addressed using Cloud Firestore. Secure user registration and login were implemented using Firebase Authentication. Another challenge was comparing lost and found items, which was handled using a weighted matching algorithm based on item name, description, category, location, and date. Handling item images was improved by adding image selection and preview functionality. Input validation and error handling were also implemented to prevent incomplete submissions and manage database or network failures.

Future Scope

The Campus Lost & Found Management System can be improved further with more advanced features in the future. The application can support better image-based matching to identify similar lost and found items more accurately. Real-time notifications can be added to immediately inform users when a possible match is found.

The system can also include chat functionality between the owner and finder, online claim verification, and an improved admin dashboard for managing reports. GPS-based location tracking can help users identify where an item was lost or found. The application can later be expanded to support multiple colleges or universities, making it a larger centralized lost and found platform.

Hardware and Software requirements

Hardware Requirements:

- Processor: Intel Core i3 or equivalent and above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free space
- Display: 1366 × 768 resolution or higher
- Android Smartphone: Android 7.0 (API 24) or above
- Internet Connection: Required for Firebase services
- Camera: Optional, for capturing item photos

Software Requirements:

- Operating System: Windows 10/11
- Android Studio: Latest compatible version
- Programming Language: Kotlin
- UI Framework: Jetpack Compose
- Backend: Firebase
- Database: Cloud Firestore
- Authentication: Firebase Authentication
- Notifications: Firebase Cloud Messaging (FCM)
- Image Processing: Google ML Kit
- Build Tool: Gradle
- Testing: Android Emulator / Android device
- Version Control: Git and GitHub (optional)

Conclusion

The Campus Lost & Found Management System provides a simple and organized way for students to report and find lost or found items on campus. The application allows users to register, log in, submit lost and found item reports, add item details and photos, and view possible matching items. Firebase is used to securely manage users and store item information. Overall, the system reduces manual work and makes the process of recovering lost belongings easier and faster.

References

- **Firestore Documentation – Firestore Authentication, Cloud Firestore, Cloud Storage and Firestore Cloud Messaging.**
- **<https://firebase.google.com/docs>**
- **Jetpack Compose Documentation – Android UI development using Kotlin and Jetpack Compose.**
- **<https://developer.android.com/compose>**
- **Android Developers Documentation – Android application development, lifecycle and UI components.**
- **<https://developer.android.com/>**
- **Google ML Kit Documentation – Image labeling and machine-learning features for mobile applications.**
- **<https://developers.google.com/ml-kit>**
- **Kotlin Documentation – Kotlin programming language and coroutines.**
- **<https://kotlinlang.org/docs/>**
- **Jaccard Similarity – Technique used for comparing similarity between sets of words.**
- **https://en.wikipedia.org/wiki/Jaccard_index**
- **Research Papers / Articles on Lost and Found Systems – Used for understanding existing approaches to item reporting, searching, and matching.**

Thank you